

```
#!/usr/bin/env python3
```

```
"""
```

```
AESOP XI JSONL Marker Generator — Termux Native & Local Node  
Reads markdown files from vault, extracts metadata and headers,  
and outputs standardized universal-index.jsonl markers.
```

```
"""
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```
import hashlib
```

```
import json
```

```
import os
```

```
import sys
```

```
from datetime import datetime
```

```
from pathlib import Path
```

```
def extract_metadata(file_path: Path) -> dict:
```

```
    content = file_path.read_text(encoding="utf-8", errors="ignore")
```

```
    title = file_path.stem
```

```
    headers = [line.strip("# ").strip() for line in content.splitlines() if line.startswith("#")]
```

```
    sha256 = hashlib.sha256(content.encode("utf-8")).hexdigest()
```

```
    return {
```

```
        "record_id": f"REC_{title.upper()[:16]}_{sha256[:8]}",
```

```
        "file_name": file_path.name,
```

```
        "path": str(file_path),
```

```
        "sha256": sha256,
```

```
        "title": headers[0] if headers else title,
```

```
        "section_count": len(headers),
```

```
        "file_size": len(content),
```

```
        "timestamp": datetime.utcnow().isoformat() + "Z"
```

```
    }
```

```
def process_directory(source_dir: str, output_file: str):
```

```
    p = Path(os.path.expanduser(source_dir))
```

```
    out = Path(os.path.expanduser(output_file))
```

```
    out.parent.mkdir(parents=True, exist_ok=True)
```

```
    with open(out, "w", encoding="utf-8") as f:
```

```
        for md_file in p.glob("**/*.md"):
```

```
            meta = extract_metadata(md_file)
```

```
            f.write(json.dumps(meta) + "\n")
```

```
    print(f"[✓] Generated JSONL markers in: {out}")
```

```
if __name__ == "__main__":
```

```
    src = sys.argv[1] if len(sys.argv) > 1 else ""
```

```
dest = sys.argv[2] if len(sys.argv) > 2 else "universal-index.jsonl"  
process_directory(src, dest)
```